Rohan Kumar Pathak

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Professional Summary

Computer Science undergraduate and aspiring Software Developer with hands-on experience in full-stack MERN development, RESTful API design, authentication, real-time applications, and Generative AI. Experienced in building responsive web applications and integrating Retrieval-Augmented Generation (RAG), embeddings, and vector search into AI-powered learning tools. Strong foundation in Data Structures and Algorithms, OOP, DBMS, and Operating Systems.

Education

IEC College of Engineering and Technology, Greater Noida **2023 – 2027**
B.Tech. in Computer Science and Engineering *CGPA: 7.0/10*

Professional Experience

The Developer Arena **July 2025 - January 2026**
Full Stack Developer Intern

- Developed reusable React.js components and implemented search-driven interfaces to streamline user interactions and improve frontend usability.
- Built and integrated RESTful APIs using Node.js and Express.js, enabling seamless communication between React frontend modules and backend services.
- Implemented backend functionality, including request validation, error handling, and database operations, to support reliable application workflows.
- Integrated Large Language Model (LLM) APIs with backend services to enable AI-powered application features and intelligent user interactions.

Projects

StudyMate AI | *React.js, Node.js, Python, FastAPI, LangChain, Hugging Face, RAG, FAISS* GitHub | Live Demo

- Built an AI-powered study assistant that transforms uploaded learning materials into context-aware answers, quizzes, and revision flashcards using Retrieval-Augmented Generation (RAG).
- Implemented document ingestion, text chunking, embedding generation, and FAISS vector similarity search to retrieve relevant context for AI-generated responses.
- Developed a FastAPI backend and responsive React interface for document uploads, interactive question answering, and personalized study workflows.

LifeLedger | *React.js, Node.js, Express.js, MongoDB, JWT, REST APIs* GitHub | Live Demo

- Engineered and deployed a MERN-based personal inventory platform for managing belongings, important documents, private collections, and essential life events.
- Implemented authentication-protected user data, item and document management, and MongoDB-backed CRUD operations using Express.js APIs and Mongoose models.
- Built warranty and expiry tracking, calendar integration, deadline notifications, and responsive light/dark themes to help users organize assets and avoid missed dates.

Real-Time Chat Application | *React.js, Node.js, Express.js, MongoDB, JWT, Socket.IO* GitHub | Live Demo

- Developed a real-time messaging application using the MERN stack and Socket.IO to enable low-latency, bidirectional communication between connected users.
- Integrated JWT-based authentication and protected user access, with MongoDB and Mongoose for user data and persistent chat history.
- Connected React frontend components with backend APIs and real-time socket events to deliver responsive messaging and seamless client-server updates.

Technical Skills

Languages: C++, JavaScript, Python, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, Axios, Responsive Design

Backend: Node.js, Express.js, FastAPI, RESTful APIs, Socket.IO

Databases: MongoDB, Mongoose, MySQL

AI / GenAI: LangChain, Hugging Face, RAG, FAISS, Embeddings, Prompt Engineering, Streamlit

Tools: Git, GitHub, Postman, VS Code, Docker

Core Concepts: DSA, OOP, DBMS, Operating Systems, JWT Authentication, Role-Based Authorization

Achievements

- Solved 150+ LeetCode problems, strengthening data structures, algorithms, and problem-solving skills.
- Secured a Top 3 position in a GeeksforGeeks hackathon for a real-time forecasting solution.

Certifications

C++ Programming – Great Learning; JavaScript – Infosys Springboard; MySQL – SkillNation; HTML & CSS – Great Learning; AI Tools Workshop – Be10x; ADCA – CertiPort Academy; GeeksforGeeks Hackfest.